The pxCORE package

Core aggregation and key management for P3CTeX

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Abstract

pxCORE is the central aggregation package in the P3CTeX stack. It owns the pxCORE key family, which controls activation of standalone libraries (pxPRP, pxUML, pxSRC, ...) and offers a minimal LaTeX2e configuration API.

Contents

1 Loading the package

In standalone documents, load pxCORE as a normal package:

```
\usepackage[<options>]{pxCORE}
```

In P3CTeX-based workflows, pxCORE is normally loaded by the document class (P3CTeX.cls) and configured via class options.

2 Readable tool selection

```
\usepackage[code,images,tables,diagrams,calculations,project-management]{pxCORE}
```

The keywords code, images, tables and diagrams select the existing LST, SRC, TAB and CMP tools. diagrams-config forwards diagram options, just like CMPconfig. calculations (or CALC) loads pxCALC; project-management (or PM) loads pxPM. The diagram entry point pxDIAG delegates to pxCMP. All selection keys accept =false; a bare keyword means true. See the student-tools guide for examples and customisation.

3 Key family pxCORE

The pxCORE key family controls which libraries are activated and stores a few global flags:

debug (bool) enables internal debugging. When set, \pxCOREIfDebugTF will take its "true" branch.

print-doc (bool) reserved flag for future "print documentation" workflows. Currently stored but not interpreted.

PRP (bool) toggles loading of the pxPRP property-map library.

UML (bool) toggles loading of the pxUML UML diagram library.

UMLconfig (token list) raw option string passed to pxUML. When nonempty, pxCORE executes

```
\RequirePackage[<UMLconfig>]{pxUML}
```

instead of a bare \RequirePackage{pxUML}.

SRC (bool) toggles loading of the pxSRC included-sources library.

Keys are typically set via package options:

```
\usepackage[PRP,UML,SRC]{pxCORE}
```

or adjusted later with \pxCOREsetup (see below).

4 User command reference

4.1 \pxCOREsetup

Synopsis

```
\pxCOREsetup{<key>=<value>,...}
```

Updates the pxCORE key family after package loading. This is useful to toggle flags such as debug programmatically or to adjust options in local scopes.

Example:

```
\pxCOREsetup{debug=true}
...
\pxCOREsetup{debug=false}
```

4.2 \pxCOREIfDebugTF

Synopsis

```
\pxCOREIfDebugTF{<true-code>}{<false-code>}
```

Branch on the `debug` flag in the `pxCORE` key family. Both branches are arbitrary LaTeX code.

Example:

```
\pxCOREIfDebugTF
  {\typeout{[pxCORE] Debug mode enabled}}
  {}
```

5 Minimal example

```
\usepackage[PRP,UML,SRC]{pxCORE}
\begin{document}
\pxCOREsetup{debug=true}
\pxCOREIfDebugTF{[DEBUG-ON]}{[DEBUG-OFF]}
\end{document}
```